

K13-A, Netherlands

WMO: 062520

Lat: 53.218N

Lon: 3.220E

Elev: 30

StdP: 100.97

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.1	1.1	-6.0	2.3	1.4	-4.3	2.6	2.2	20.3	6.9	18.9	7.1	8.9	90	0.836

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.8	20.0	18.0	19.2	17.4	18.7	17.0	18.4	19.6	17.8	19.0	17.3	18.5	6.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	12.9	19.2	17.2	12.3	18.7	16.7	12.0	18.3	52.2	19.8	50.3	19.1	48.8	18.4	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	16.5	15.0	-0.6	21.4	2.1	1.1	-2.1	22.2	-3.4	22.8	-4.6	23.4	-6.2	24.2		
(5)			-2.4	19.0	2.3	0.9	-4.1	19.6	-5.4	20.1	-6.7	20.7	-8.4	21.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	6.2	5.5	6.2	8.0	10.7	13.4	15.9	17.1	16.1	13.4	10.2	7.7		
	DBStd	4.46	2.28	2.28	1.97	1.79	1.71	1.56	1.49	1.34	1.57	2.07	2.10	2.30		
	HDD10.0	536	117	125	118	63	13	0	0	0	0	2	21	76		
	HDD18.3	2719	375	358	377	309	238	149	76	42	69	153	243	329		
	CDD10.0	867	0	0	0	4	33	102	184	222	182	107	28	4		
	CDD18.3	8	0	0	0	0	0	0	2	4	2	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH27.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	7.9	9.4	8.7	8.3	7.3	7.4	6.7	6.6	6.6	7.3	8.7	8.8	9.3		
(15) Precipitation	PrecAvg	975	94	72	66	46	52	63	74	93	95	104	108	108		
	PrecMax	1262	165	146	139	99	88	135	172	209	192	191	192	182		
	PrecMin	724	12	23	10	3	24	19	13	20	19	37	18	50		
	PrecStd	132	39	31	31	24	19	30	36	46	43	43	39	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	10.3	10.6	13.0	15.4	18.7	20.9	21.4	20.8	18.2	15.8	13.1		
		MCWB	9.8	8.6	8.9	10.5	13.0	16.2	18.2	18.6	18.5	16.2	14.1	11.5		
	2%	DB	10.5	9.6	9.7	11.6	14.2	16.9	19.4	20.3	19.2	17.2	14.9	12.2		
		MCWB	9.3	8.2	8.4	9.9	12.4	15.2	17.6	18.3	17.3	15.6	13.3	10.7		
	5%	DB	9.8	9.0	9.1	10.9	13.5	15.9	18.7	19.5	18.8	16.7	13.9	11.6		
		MCWB	8.6	7.8	7.9	9.4	11.9	14.5	17.1	17.8	17.0	15.1	12.2	10.1		
	10%	DB	9.1	8.4	8.4	10.2	12.9	15.2	18.0	18.9	18.2	16.1	13.1	10.7		
		MCWB	7.9	7.2	7.3	9.0	11.5	13.9	16.5	17.3	16.3	14.4	11.2	9.2		
	0.4%	WB	10.3	9.2	9.3	11.0	13.7	16.6	18.7	19.5	18.7	16.6	14.6	12.0		
		MCDB	11.0	9.7	10.0	12.2	14.7	18.0	20.2	20.3	20.0	17.7	15.7	12.7		
	2%	WB	9.5	8.5	8.6	10.2	12.8	15.5	18.0	18.7	17.7	16.0	13.5	11.2		
		MCDB	10.3	9.2	9.4	11.2	13.8	16.4	19.2	19.8	18.9	17.1	14.6	12.2		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	8.8	8.0	8.1	9.6	12.2	14.8	17.3	18.1	17.4	15.4	12.5	10.3		
		MCDB	9.6	8.8	8.8	10.6	13.2	15.7	18.5	19.3	18.5	16.5	13.8	11.4		
	10%	WB	8.0	7.3	7.5	9.1	11.7	14.2	16.7	17.5	16.8	14.8	11.5	9.5		
		MCDB	8.9	8.2	8.2	10.1	12.7	15.1	17.8	18.8	18.1	16.0	12.9	10.6		
	5% DB	MDBR	2.2	2.0	2.0	2.0	1.9	1.8	1.9	1.8	1.9	2.2	2.2	2.3		
		MCDBR	2.7	2.3	2.3	2.7	2.7	2.7	2.7	2.3	2.2	2.5	2.7	2.6		
(36) Mean Daily Temperature Range	5% WB	MCWBR	3.2	2.3	2.2	1.9	2.0	2.3	1.9	2.1	2.4	2.9	3.2	3.2		
		MCDBR	2.6	2.2	2.2	2.4	2.6	2.7	2.5	2.1	2.1	2.5	2.6	2.5		
	5% WB	MCWBR	3.2	2.4	2.2	1.9	2.2	2.4	2.0	2.1	2.4	3.0	3.2	3.2		
		MCDBR	2.6	2.2	2.2	2.4	2.6	2.7	2.5	2.1	2.1	2.5	2.6	2.5		
(40) Clear-Sky Solar Irradiance	taub		0.332	0.348	0.377	0.402	0.400	0.403	0.408	0.398	0.385	0.363	0.338	0.322		
	taud		2.326	2.364	2.312	2.281	2.362	2.366	2.361	2.401	2.429	2.424	2.366	2.314		
	Ebn at Noon		632	749	800	827	850	849	837	825	789	722	620	567		
	Edh at Noon		61	79	101	118	115	115	114	104	90	74	58	52		
(44) All-Sky Solar Radiation	RadAvg		0.64	1.37	2.70	4.46	5.55	5.81	5.61	4.71	3.26	1.75	0.78	0.48		
	RadStd		0.06	0.23	0.23	0.42	0.36	0.31	0.58	0.33	0.24	0.14	0.07	0.05		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page